CLASS PRESENTATION

AGILE PROJECT MANAGEMENT

Plan small. Deliver often. Improve continuously.



Agile • Scrum • Sprints • Product Backlog

Presented by: _____

Today’s roadmap

What we will understand by the end

01

Agile basics

Meaning, goal and mindset

02

How it works

Short cycles, testing and feedback

03

Scrum

Roles, events and Sprint

04

Backlogs

Product Backlog vs Sprint Backlog

05

Example

Apply Agile to a college system

Key question: How can a team deliver useful software while requirements keep changing?

1. What is Agile?

A flexible way to develop and deliver a product

Agile is an iterative approach where a team builds a product in small parts, tests each part, gathers feedback and adapts as needed.

SMALL BATCHES | FAST FEEDBACK | ADAPT TO CHANGE

Agile is a mindset guided by values and principles—not one single tool.

Think of it this way

Instead of waiting months for one big release, deliver useful features step by step.

2. Why use Agile?

Agile helps teams respond to real user needs

Flexibility

Requirements can be reprioritized when needs change.

Early value

Users can try useful features before the entire product is finished.

Frequent feedback

Problems and misunderstandings can be found earlier.

Team learning

Each cycle helps the team improve its product and process.

Agile does not mean “no planning”; it means planning continuously as the team learns.

3. Four values of the Agile Manifesto

The values behind Agile ways of working

People & interaction

Team communication matters more than relying only on processes.

Working software

A working product is more valuable than documentation alone.

Customer collaboration

Work with the customer, not just around a contract.

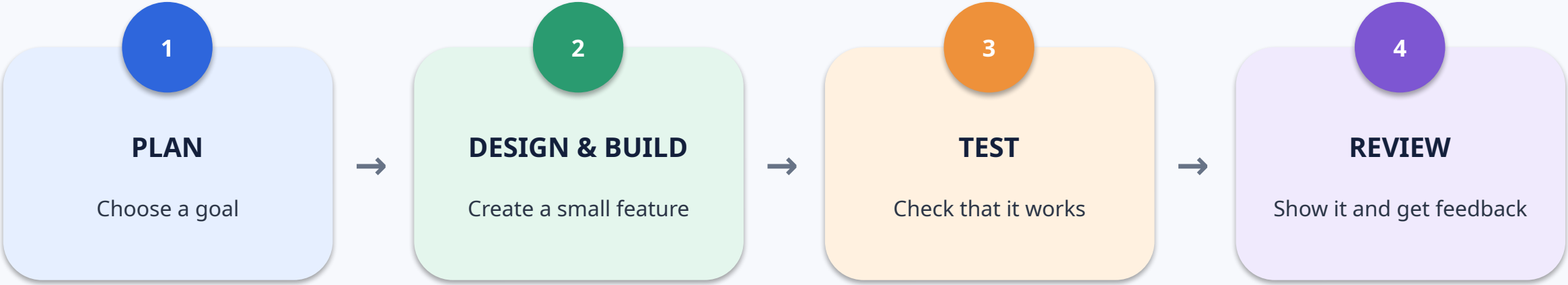
Respond to change

Adapt to new information instead of rigidly following an old plan.

The Agile Manifesto values the left-hand items, while still recognizing value in the right-hand items.

4. Agile workflow

A repeatable cycle of planning, building and learning



LEARN FROM FEEDBACK → ADAPT THE PLAN → REPEAT

In Scrum, this fixed-length cycle is called a Sprint.

5. Agile vs Waterfall

Two different ways to organize project work

AGILE	WATERFALL
✓ Work in short iterations	• Phases generally follow a sequence
✓ Frequent feedback	• Feedback often at milestones
✓ Changes can be accommodated	• Late changes can be costly
✓ Working features delivered regularly	• Delivery often comes after major phases

Neither approach is best for every situation; choose based on the project and its constraints.

6. Scrum: a popular Agile framework

Scrum gives teams a clear structure for working in Sprints

Product Owner

Orders Product Backlog items and focuses on product value.

Scrum Master

Helps everyone understand Scrum and improve effectiveness.

Developers

Build a usable product increment and manage their work.

Scrum = roles/accountabilities + events + artifacts + clear goals

Note: Scrum uses the term “Developers” for the people creating the product; they may have different specialties.

7. Sprint and Scrum events

Events help the team plan, inspect and improve

1. **Sprint Planning** — Agree on why the Sprint matters and what work is realistic.
2. **Daily Scrum** — Developers inspect progress toward the Sprint Goal and adapt their plan.
3. **Sprint Review** — Inspect the outcome with stakeholders and discuss what to do next.
4. **Sprint Retrospective** — Identify improvements to quality, collaboration and ways of working.

A Sprint is one month or less; many teams use one- or two-week Sprints.

8. Product Backlog: the big picture

An ordered list of everything the product may need

COLLEGE MANAGEMENT SYSTEM — EXAMPLE

1	Student login	High
2	Attendance module	High
3	Result management	Medium
4	Notice & events	Medium
5	Dark mode	Low

Who orders it?

The Product Owner is accountable for ordering the backlog.

Does it stay fixed?

No. It is refined and reordered as the team learns.

What can it contain?

Features, improvements, defects and technical work.

Priority labels are illustrative; real ordering depends on value, risk, dependencies and feedback.

9. Product Backlog vs Sprint Backlog

A common presentation and viva question

PRODUCT BACKLOG

- Ordered list for the product
- Can include future features and fixes
- Product Owner is accountable for ordering
- Evolves as needs and priorities change

SPRINT BACKLOG

- Sprint Goal + selected backlog items + delivery plan
- Focuses on the current Sprint
- Created and adapted by Developers
- Updated as the team learns during the Sprint

Product Backlog → **Sprint Planning** → **Sprint Backlog** → **Product Increment**

Easy memory trick: Product Backlog = whole product to-do list; Sprint Backlog = current Sprint plan.

10. What does a backlog item look like?

A user story explains the need from the user’s point of view

USER STORY

“As a student, I want to log in securely so that I can view my attendance.”

Acceptance criteria

Clear checks that show when the feature is complete.

Example acceptance criteria

- A registered student can log in with valid credentials.
- Invalid credentials show a helpful error message.
- After login, the student can access the attendance page.

Good backlog items are understandable, valuable, ordered and refined enough for the team to work on.

11. Apply Agile: College Management System

Example of delivering the product in stages

SPRINT 1

Student login & profile

Build • Test • Review

SPRINT 2

Attendance management

Build • Test • Review

SPRINT 3

Result management

Build • Test • Review

SPRINT 4

Notices & notifications

Build • Test • Review

After each Sprint: collect feedback, fix issues and reprioritize the backlog.

This is an illustrative plan; Sprint scope depends on team size, complexity and capacity.

12. Advantages and limitations

Agile works best when the team collaborates and stays disciplined

ADVANTAGES

- Useful features can arrive earlier
- Feedback can reveal issues sooner
- Easier to adapt to changing needs
- Team reflects and improves regularly

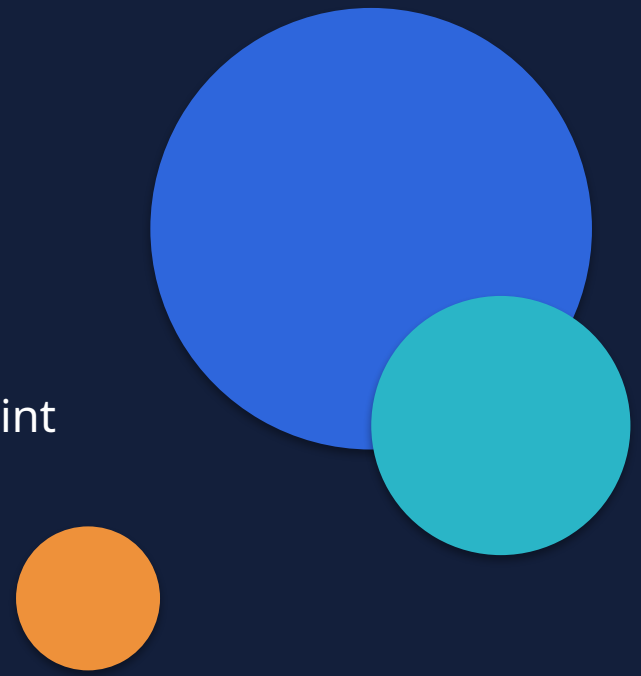
LIMITATIONS

- Needs frequent communication
- Scope may expand without control
- Customer/stakeholder availability matters
- Time and cost estimates can change

Agile is not “change everything anytime”—changes still need prioritization and clear goals.

KEY TAKEAWAYS

- Agile delivers work in small, manageable cycles.
- Scrum is one popular framework for applying Agile.
- Product Backlog lists product needs; Sprint Backlog plans current Sprint work.
- Feedback and continuous improvement are central.



THANK YOU!

Questions?